



Network Infrastructure Architecture Report

Tech Solutions (SARL)

Version 1.0 — January 20, 2026

Executive Summary

This document provides a complete professional overview of the network infrastructure designed for Tech Solutions (SARL). The architecture simulates a modern enterprise network using GNS3, featuring routed backbone segments, VLAN segmentation, firewall policies, and Linux-based services.

The objective is to model a scalable, secure, and realistic environment for engineering, training, and testing purposes. It provides system engineers and network architects with a clear understanding of the network's logical and physical design, implementation steps, and operational considerations.

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1 Introduction

The Tech Solutions (SARL) network architecture project is a fully simulated enterprise environment developed within GNS3. It includes multi-router infrastructure, departmental VLANs, dynamic routing, and server resources.

The purpose of this report is to present:

- network design principles,
- device roles and interconnections,
- used technologies and protocols,
- deployment and implementation steps,
- configuration and testing methodology.

1.1 Departmental Allocation and Logical Topology

After the integration of the Internet service, the ISP allocated the address space **172.24.0.0/14** to the enterprise. The logical topology and IP allocation for each department are summarized below:

Department	Local Router	Service / VM	PC Clients	Role in the Enterprise
Web / Marketing	RZ-1	Web Server	8905	Manages the website and customer-facing interactions
Supervision / IT	RZ-2	Monitoring Server	1465	Monitors server and network performance
Base de données / Gestion	RZ-3	Database Server	489	Centralizes and secures client and internal data
Partage / Collaboration	RZ-4	NFS Server	165	Enables internal file and document sharing

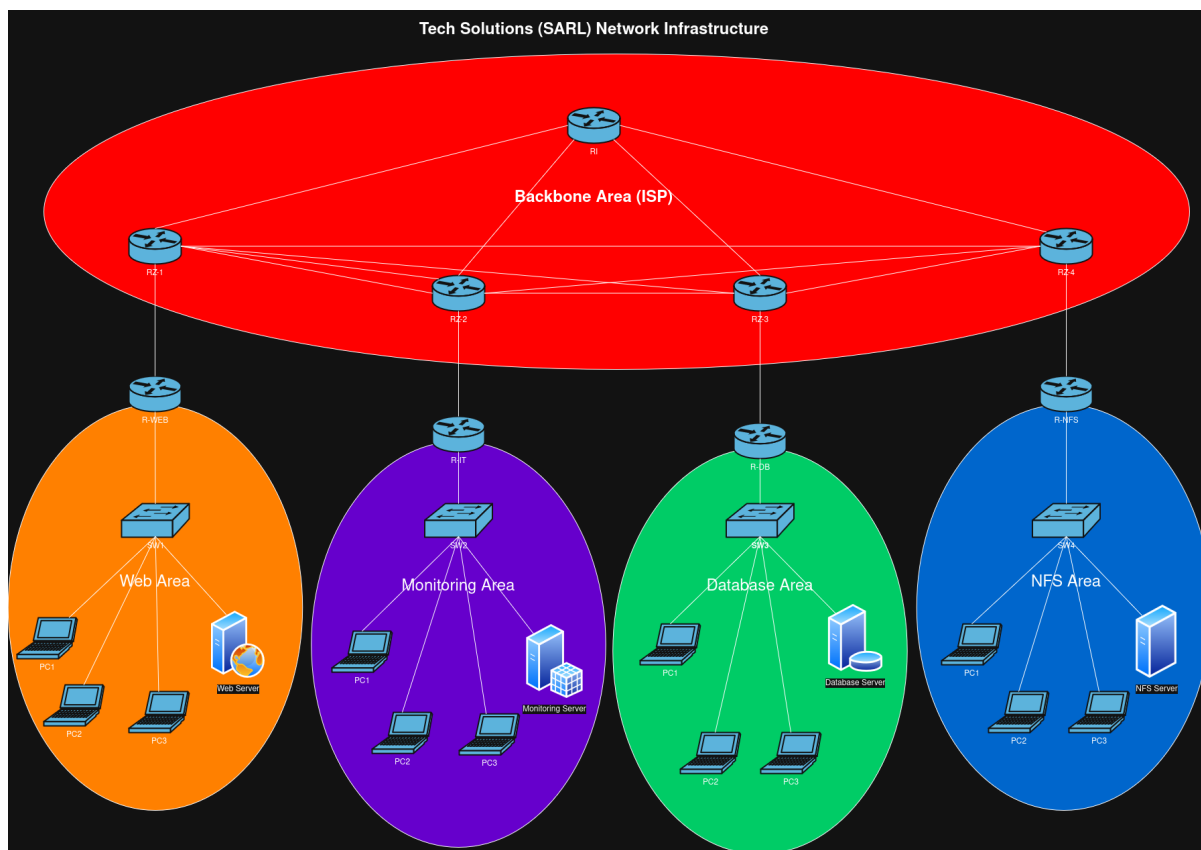
1.2 Subnetting (VLSM) Diagram

The network 172.24.0.0/14 has 262142 hosts. Your subnets need 11032 hosts.								
NAME	HOSTS NEEDED	HOSTS AVAILABLE	UNUSED HOSTS	NETWORK ADDRESS	SLASH	MASK	USABLE RANGE	BROADCAST
Web	8907	16382	7475	172.24.0.0	/18	255.255.192.0	172.24.0.1 - 172.24.63.254	172.24.63.255
IT	1467	2046	579	172.24.64.0	/21	255.255.248.0	172.24.64.1 - 172.24.71.254	172.24.71.255
DB	491	510	19	172.24.72.0	/23	255.255.254.0	172.24.72.1 - 172.24.73.254	172.24.73.255
NFS	167	254	87	172.24.74.0	/24	255.255.255.0	172.24.74.1 - 172.24.74.254	172.24.74.255

2 Technologies Used

- **GNS3 3.0.5** — Network simulation environment.
- **Cisco Routers (3725/3745)** — Core routing infrastructure.
- **Ubuntu Server 20.04** — Server and network service hosting.
- **OSPF** — Dynamic routing protocol.
- **VLANs** — Logical segmentation and broadcast containment.
- **Nmap & Wireshark** — Analysis and testing tools.
- **ZeroTier** — Potential remote access overlay network.
- **SSH / Telnet** — Device management protocols.

3 Architecture Design



3.1 Core Layer

The backbone contains the primary routing equipment enabling communication between all segments.

3.2 Distribution Layer

VLAN segmentation and policy enforcement occur at this layer.

3.3 Access Layer

End-user devices and servers connect at this level.

4 Implementation Steps

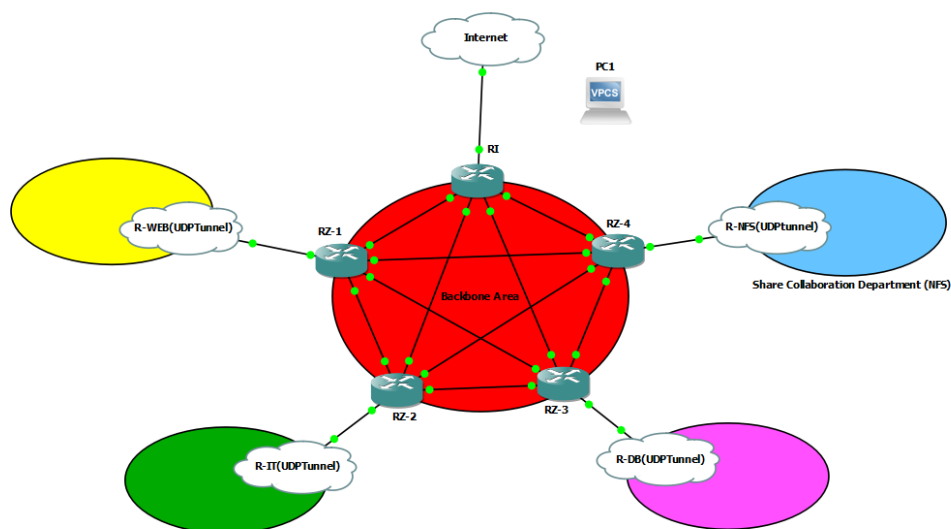
1. Prepare GNS3 environment and import appliance templates.
2. Deploy routers, switches, and servers.
3. Configure VLANs and interfaces.
4. Configure OSPF across backbone routers.
5. Implement ACLs and security policies.
6. Verify connectivity and routing.
7. Test bandwidth, latency, and redundancy.

5 Security Model

Security policies were applied on multiple layers:

- ACLs restricting cross-department traffic.
- VLAN isolation.
- SSH-enabled secure device management.
- Logging and monitoring using Linux servers.

6 Backbone Infrastructure



6.1 Purpose

The backbone provides high-availability routed connectivity between all departments, the Internet, and security services. It ensures redundancy, dynamic routing, and controlled external access.

6.2 Architecture Overview

The backbone consists of:

- Four core routers (CR1–CR4)
- OSPF dynamic routing
- FortiGate firewall (VPN + filtering)
- Internet edge router with NAT

6.3 OSPF Design

- OSPF Area 0 (Backbone)
- All core routers participate in Area 0
- Point-to-point links between core routers
- Route summarization applied toward distribution routers

6.4 VPN and Security

- FortiGate provides site-to-site and remote-access VPN

- Encrypted tunnels secure management and inter-site traffic
- Firewall policies restrict inbound/outbound traffic

6.5 Internet Router (NAT)

- Static default route toward ISP
- PAT (NAT overload) for internal subnets
- Controlled port forwarding for public services

6.6 Configuration

6.6.1 Backbone routers configuration

```
1 ! Gl3diator <Ghaith.BenSalah@esprit.tn>
2 ! Since:          2025-12-01
3 ! LastUpdate:     2026-01-20
4 ! RouterName:     R-*
5 ! RouterVersion:  C3745
6 ! See             https://github.com/Gl3diator/Projet-Systeme-et-Reseau-GNS3
7 ! COPY-THAN-PASTE
8
9 conf t
10
11 ! R-internet Backbone Interface Configuration
12 interface FastEthernet0/0
13     ip address 200.200.20.1 255.255.255.252
14     no shutdown
15
16 interface FastEthernet0/1
17     ip address 200.200.20.5 255.255.255.252
18     no shutdown
19
20 interface FastEthernet1/0
21     ip address 200.200.20.9 255.255.255.252
22     no shutdown
23
24 interface FastEthernet2/0
25     ip address 200.200.20.13 255.255.255.252
26     no shutdown
27
28 ! OSPF Routing (Area 0)
29 ! Example (RZ-4)
30 router ospf 1
31     router-id 5.5.5.5
32     network 200.200.20.0 0.0.0.63 area 0
33
34 ! Internet Access via RI (NAT)
35 ! Internet Interface (Cloud to RI)
```

```
36 interface FastEthernet3/0
37     ip address 192.168.40.200 255.255.255.0
38     ip nat outside
39     no shutdown
40
41 ! NAT Inside
42 interface FastEthernet0/0
43     ip nat inside
44 interface FastEthernet0/1
45     ip nat inside
46 interface FastEthernet1/0
47     ip nat inside
48 interface FastEthernet2/0
49     ip nat inside
50
51 ! NAT ACL
52 access-list 1 permit 200.200.20.0 0.0.0.63
53 ip nat inside source list 1 interface FastEthernet3/0 overload
54
55 ! Default Route
56 ip route 0.0.0.0 0.0.0.0 192.168.40.2
57
58 ! DNS on All Routers
59 ip domain-lookup
60 ip name-server 8.8.8.8
61 ip name-server 1.1.1.1
```

Listing 1: Backbone Routers Configuration

6.6.2 NFS router VPN configuration

```
1 ! ZouariOmar <zouariomar20@gmail.com>
2 ! Since: 2026-01-08
3 ! LastUpdate: 2026-01-08
4 ! RouterName: R-nfs
5 ! RouterVersion: C3745
6 ! COPY-THAN-PASTE
7
8 ! === IKE / ISAKMP ===
9
10 crypto isakmp policy 10
11     encr aes 256
12     hash sha
13     authentication pre-share
14     group 2
15     lifetime 86400
16
17 ! Pre-Shared Key
18 crypto isakmp key VPNKEY address 10.0.0.1
19
20 ! === IPsec ===
```

```
21
22 ! Transform Set
23 crypto ipsec transform-set TS esp-aes 256 esp-sha-hmac
24 mode tunnel
25
26 ! Interesting Traffic (ACL)
27 ip access-list extended VPN-TRAFFIC
28 permit ip 172.24.74.0 0.0.0.255 10.0.0.0 0.0.0.255
29
30 ! Crypto Map
31 crypto map CMAP 10 ipsec-isakmp
32 set peer 10.0.0.1
33 set transform-set TS
34 match address VPN-TRAFFIC
35
36 ! Apply Crypto Map
37 interface FastEthernet0/1
38 crypto map CMAP
39
40 ! Verification Commands
41 show crypto isakmp sa
42 show crypto ipsec sa
43 show access-lists VPN-TRAFFIC
44 ! Example trigger
45 ping 172.24.74.1 source 10.0.0.1
```

Listing 2: R-NFS VPN configuration

6.6.3 Backbone router VPN configuration

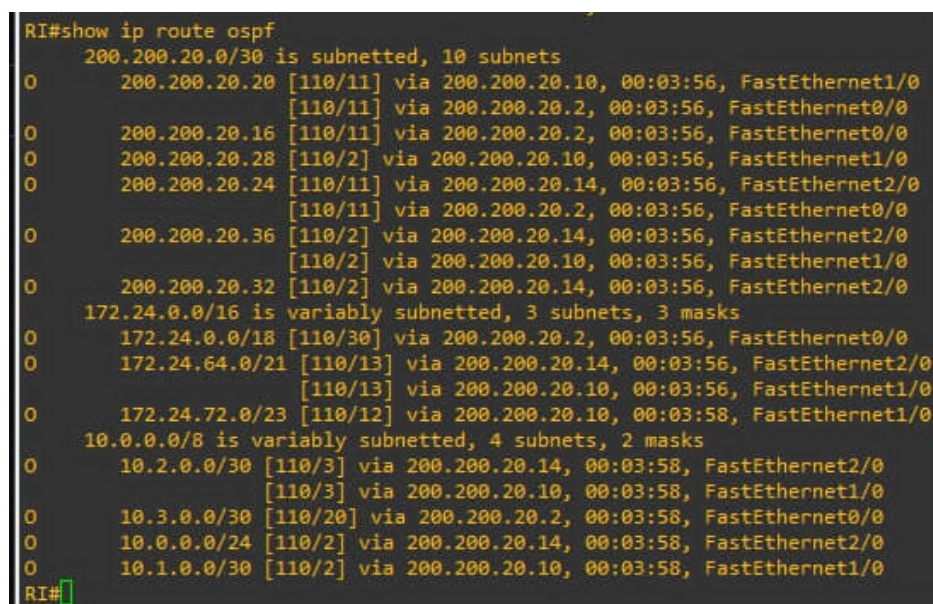
```
1 ! G13diator <ghaithbensalah1999@gmail.com>
2 ! Since: 2026-01-08
3 ! LastUpdate: 2026-01-08
4 ! RouterName: R-z4
5 ! RouterVersion: C3745
6 ! COPY-THAN-PASTE
7
8 ! === IKE / ISAKMP ===
9
10 crypto isakmp policy 10
11 encr aes 256
12 hash sha
13 authentication pre-share
14 group 2
15 lifetime 86400
16
17 ! Pre-Shared Key
18 crypto isakmp key VPNKEY address 10.0.0.2
19
20 ! === IPsec ===
21
```

```
22 ! Transform Set
23 crypto ipsec transform-set TS esp-aes 256 esp-sha-hmac
24 mode tunnel
25
26 ! Interesting Traffic (ACL)
27 ip access-list extended VPN-TRAFFIC
28 permit ip 10.0.0.0 0.0.0.255 172.24.74.0 0.0.0.255
29
30 ! Crypto Map
31 crypto map CMAP 10 ipsec-isakmp
32 set peer 10.0.0.2
33 set transform-set TS
34 match address VPN-TRAFFIC
35
36 ! Apply Crypto Map
37 interface FastEthernet1/0
38 crypto map CMAP
39
40 ! Verification Commands
41 show crypto isakmp sa
42 show crypto ipsec sa
43 show access-lists VPN-TRAFFIC
```

Listing 3: RZ4 VPN configuration

6.7 Screenshots

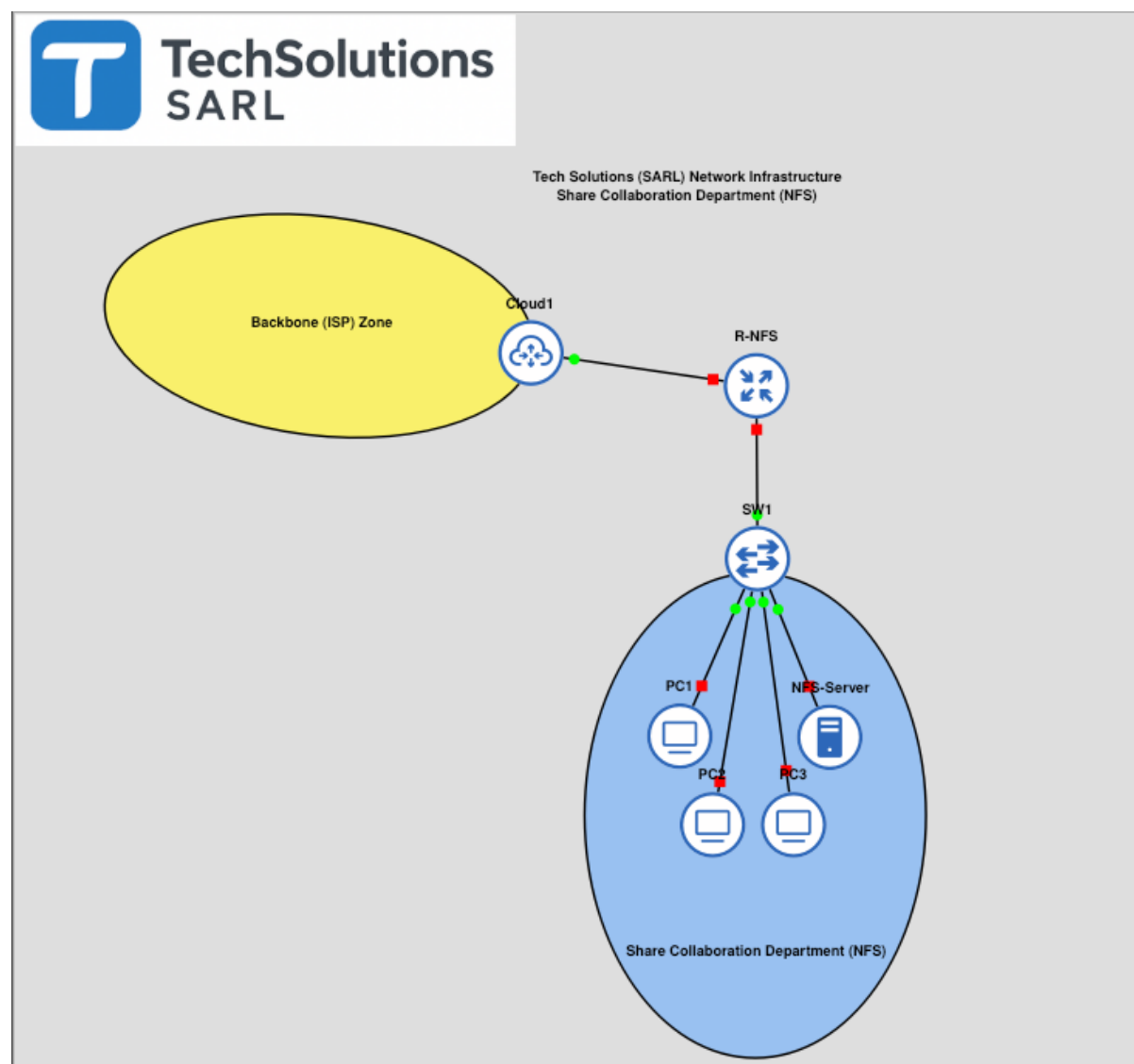
6.7.1 Backbone - OSPF Protocol



```
RI#show ip route ospf
 200.200.20.0/30 is subnetted, 10 subnets
O    200.200.20.20 [110/11] via 200.200.20.10, 00:03:56, FastEthernet1/0
O    [110/11] via 200.200.20.2, 00:03:56, FastEthernet0/0
O    200.200.20.16 [110/11] via 200.200.20.2, 00:03:56, FastEthernet0/0
O    200.200.20.28 [110/2] via 200.200.20.10, 00:03:56, FastEthernet1/0
O    200.200.20.24 [110/11] via 200.200.20.14, 00:03:56, FastEthernet2/0
O    [110/11] via 200.200.20.2, 00:03:56, FastEthernet0/0
O    200.200.20.36 [110/2] via 200.200.20.14, 00:03:56, FastEthernet2/0
O    [110/2] via 200.200.20.10, 00:03:56, FastEthernet1/0
O    200.200.20.32 [110/2] via 200.200.20.14, 00:03:56, FastEthernet2/0
 172.24.0.0/16 is variably subnetted, 3 subnets, 3 masks
O    172.24.0.0/18 [110/30] via 200.200.20.2, 00:03:56, FastEthernet0/0
O    172.24.64.0/21 [110/13] via 200.200.20.14, 00:03:56, FastEthernet2/0
O    [110/13] via 200.200.20.10, 00:03:56, FastEthernet1/0
O    172.24.72.0/23 [110/12] via 200.200.20.10, 00:03:58, FastEthernet1/0
 10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
O    10.2.0.0/30 [110/3] via 200.200.20.14, 00:03:58, FastEthernet2/0
O    [110/3] via 200.200.20.10, 00:03:58, FastEthernet1/0
O    10.3.0.0/30 [110/20] via 200.200.20.2, 00:03:58, FastEthernet0/0
O    10.0.0.0/24 [110/2] via 200.200.20.14, 00:03:58, FastEthernet2/0
O    10.1.0.0/30 [110/2] via 200.200.20.10, 00:03:58, FastEthernet1/0
RI#
```

Figure 1: OSPF Active On Backbone

7 NFS Department



7.1 Purpose

This department provides centralized file sharing services for internal collaboration and data exchange.

7.2 Architecture

- Linux NFS Server (Ubuntu Server)
- Client workstations
- Dedicated router and access switch

7.3 Server Configuration Overview

- NFS service enabled

- Shared directories defined in `/etc/exports`
- Access restricted to internal subnets

7.4 Client Configuration Script Overview

Clients mount shared directories automatically at boot using:

- `mount -t nfs`
- `/etc/fstab`

7.5 Router Script Overview

- Interface IP configuration
- OSPF advertisement
- ACL allowing NFS traffic only

7.6 Switch Script Overview

- VLAN assignment
- Access ports for clients
- Trunk port toward router

7.7 Scripts and Configuration

7.7.1 NFS Router Configuration

```
1  ! ZouariOmar <zouariomar20@gmail.com>
2  ! Since:          2025-11-19
3  ! LastUpdate:     2026-01-05
4  ! RouterName:     R-nfs
5  ! RouterVersion:  C3745
6  ! COPY-THAN-PASTE
7
8  conf t
9  hostname R-nfs
10
11 ! =====
12 ! ===== INTERFACES PART =====
13 ! === FO/1(WAN) - FO/0(LAN) ===
14 ! =====
15
16 ! === FastEthernet0/1 (WAN) ===
17 interface fa0/1
18     description R-nfs to UDP Tunnel
19     ip address 10.0.0.2 255.255.255.0
20     no shutdown
21     exit
22
```

```
23 ! === FastEthernet0/0 (LAN) ===
24 interface fa0/0
25     description R-nfs to SW1
26     ip address 172.24.74.1 255.255.255.0
27     no shutdown
28     exit
29
30 ! =====
31 ! ===== SERVICES PART =====
32 ! ===== OSPF - DHCP - DNS =====
33 ! =====
34
35 ! === OSPF ===
36 router ospf 1
37     router-id 7.7.7.7
38     network 10.0.0.0 0.0.0.255 area 0
39     network 172.24.74.0 0.0.0.255 area 0
40     passive-interface default
41     no passive-interface FastEthernet0/1
42     ! NOTE: 'f0/0' should not be no 'passive-interface'
43     default-information originate
44     exit
45
46 ! Default route toward backbone
47 ip route 0.0.0.0 0.0.0.0 10.0.0.1
48
49 ! === DHCP ===
50 ip dhcp pool NFS-DHCP
51     network 172.24.74.0 255.255.255.0
52     default-router 172.24.74.1
53     dns-server 8.8.8.8
54     exit
55
56 ! Exclude @GW
57 ip dhcp excluded-address 172.24.74.1
58
59 ! Exclude @NFS-server
60 ip dhcp excluded-address 172.24.74.2
61
62 ! === DNS ===
63 ip domain-lookup
64     ip name-server 8.8.8.8
65
66 ! =====
67 ! ===== SECURITY PART =====
68 ! == CONSOLE - SSH - TELNET - ACL ==
69 ! =====
70
71 ! Create an Admin user
72 username admin privilege 15 secret <STRONG_PASSWORD>
73
```

```
74 ! Encrypt all passwords
75 service password-encryption
76
77 ! === Console ===
78 line console 0
79     login local
80     exec-timeout 5 0      ! Auto-logout after 5min
81     logging synchronous ! Clean console display
82     exit
83
84 ! === SSH ===
85 crypto key generate rsa usage-keys label SSH modulus 2048
86 ip ssh version 2
87 ! ip access-list standard SSH_ONLY
88 !     permit X.X.X.X X.X.X.X
89 !     deny any
90 line vty 0 4
91     login local
92     transport input ssh
93 ! access-class SSH_ONLY in
94     exec-timeout 5 0
95     logging synchronous
96     exit
97
98 ! === Telnet ===
99 ! ip access-list standard TELNET_ONLY
100 !     permit 10.0.0.0 0.0.0.255
101 !     deny any
102 line vty 5 15
103     login local
104     transport input telnet
105 ! access-class TELNET_ONLY in
106     exec-timeout 5 0
107     logging synchronous
108     end
109
110 ! Disable unused services
111 ! no ip http server          ! HTTP
112 ! no ip http secure-server  ! HTTPS
113 ! no cdp run                ! CDP
114
115 write
116
117 ! =====
118 ! === VERIFICATION PART ===
119 ! =====
120
121 ! === OSPF Interfaces ===
122 ! Check which interfaces run OSPF and their state
123 show ip ospf interface brief
124 show ip ospf interface
```

```
125 show ip protocols
126
127 ! === OSPF Interfaces ===
128 ! Verify adjacency with other routers (if backbone is up)
129 show ip ospf neighbor
130 show ip ospf neighbor detail
131
132 ! === OSPF Routes ===
133 ! Verify OSPF learned routes
134 show ip route ospf
135 show ip route
136
137 ! === Default Route / Redistribution ===
138 ! Check default route advertisement
139 show ip route 0.0.0.0
140 show running-config | include default-information
141
142 ! === LAN Connectivity / DHCP ===
143 ! Verify DHCP bindings and pool
144 show ip dhcp binding
145 show ip dhcp pool
146
147 ! === DNS Resolution ===
148 ! Verify DNS lookup from router
149 ping 8.8.8.8
150 ping www.google.com
```

Listing 4: NFS Router Configuration

7.7.2 NFS Server Script

```
1 #!/usr/bin/env bash
2 # =====
3 # .FILE
4 #   nfs-server-conf
5 #
6 # .SYNOPSIS
7 #   Bash script to configure NFS-Linux server
8 #
9 # .DESCRIPTION
10 #   I write this script for
11 #   'Linux nfs-server 6.8.0-88-generic #89-Ubuntu SMP
12 #   PREEMPT_DYNAMIC Sat Oct 11 01:02:46 UTC 2025 x86_64 x86_64 GNU
13 #   /Linux'
14 #   (ubuntu-24.04.3-live-server-amd64)
15 #
16 # .PARAMETER NONE
17 #   NONE
18 #
19 # .INPUTS
20 #   NONE
```

```

19 #
20 # .OUTPUTS
21 #   NONE
22 #
23 # .NOTES
24 #   Version      : 1.0
25 #   Author       : @ZouariOmar <zouariomar20@gmail.com>
26 #   Created      : 11/22/2025
27 #   Change Log   : 9780b36
28 #
29 # .EXAMPLE
30 #   > ./nfs-server-conf
31 #   --
32 #   > NET_INTERFACE="enp2s0" \
33 #     NET_ADDRESS="172.24.74.2/24" \
34 #     NET_GW="172.24.74.1" \
35 #     NET_DNS_SERVERS="[8.8.8.8]" \
36 #     SHARED_DIR="/srv/nfs/share" \
37 #     NFS_GROUP='zouari_omar'
38 #     ./nfs_server_conf
39 # =====
40
41 #####
42 ### CUSTOM CONFIGURATION PART ###
43 #####
44 NET_INTERFACE="enp2s0"
45 NET_SUBNET='X.X.X.X/XX'
46 NET_ADDRESS="X.X.X.X/XX"
47 NET_GW="X.X.X.X"
48 NET_DNS_SERVERS="[X.X.X.X]"
49 SHARED_DIR='path_to_shared_directory'
50 NFS_GROUP='group_to_chown_shared_dir'
51
52 #####
53 ### GENERAL CONFIGURATION PART ###
54 #####
55
56 # Set keymap into fr
57 sudo sed -i 's/XKBLAYOUT=".*"/XKBLAYOUT="fr"/' /etc/default/
    keyboard
58
59 # Update the system
60 sudo apt update -y && sudo apt upgrade -y
61
62 # Change the current hostname
63 hostnamectl set-hostname nfs-server
64
65 #####
66 ### NETWORK CONFIGURATION PART ###
67 #####
68

```

```
69 # Set the private static ip address
70 sudo bash -c "cat << 'EOF' > /etc/netplan/01-static.yaml
71 network:
72   version: 2
73   renderer: networkd
74   ethernets:
75     ${NET_INTERFACE}:
76       dhcp4: no
77       addresses:
78         - ${NET_ADDRESS}
79       nameservers:
80         addresses: ${NET_DNS_SERVERS}
81       routes:
82         - to: default
83           via: ${NET_GW}
84 EOF"
85
86 # Change 01-static.yaml accessibility
87 sudo chmod 600 /etc/netplan/01-static.yaml
88
89 # Apply changes
90 sudo netplan apply
91
92 #####
93 ### NFS-SERVER CONFIGURATION PART ###
94 #####
95
96 # Install Server NFS-SERVER package
97 sudo apt install -y nfs-kernel-server nfs-common
98
99 # Enable NFS service
100 sudo systemctl enable --now nfs-server
101
102 # Create and Export a Shared Directory
103 sudo mkdir -p ${SHARED_DIR}
104
105 # Add local subnet to the ACL (Access Control List)
106 echo "${SHARED_DIR} ${NET_SUBNET}(rw, sync, no_subtree_check)" |
    sudo tee -a /etc/exports
107
108 # Apply exports
109 sudo exportfs -ra
110
111 # Give grp permission
112 # NOTE: To add new group 'sudo groupadd NFS_GROUP'
113 # NOTE: To delete group 'sudo groupdel NFS_GROUP'
114 sudo chown -R :${NFS_GROUP} ${SHARED_DIR}
115 sudo chmod -R 775 ${SHARED_DIR}
116
117 # Configure Firewall
118 sudo ufw allow from ${NET_SUBNET} to any port nfs
```

Listing 5: NFS Server Configuration Script

7.7.3 NFS Client Script

```

1  #!/usr/bin/env bash
2  # =====
3  # .FILE
4  #   nfs_client_conf
5  #
6  # .SYNOPSIS
7  #   Bash script to configure NFS-Linux client
8  #
9  # .DESCRIPTION
10 #   I write this script for
11 #   'Linux nfs-server 6.8.0-88-generic #89-Ubuntu SMP
    PREEMPT_DYNAMIC Sat Oct 11 01:02:46 UTC 2025 x86_64 x86_64 GNU
    /Linux '
12 #   (ubuntu-24.04.3-live-server-amd64)
13 #
14 # .PARAMETER NONE
15 #   NONE
16 #
17 # .INPUTS
18 #   NONE
19 #
20 # .OUTPUTS
21 #   NONE
22 #
23 # .NOTES
24 #   Version      : 1.0
25 #   Author       : @ZouariOmar <zouariomar20@gmail.com>
26 #   Created      : 11/22/2025
27 #   Change Log   : 9780b36
28 #
29 # .EXAMPLE
30 #   > ./nfs-client-conf
31 #   --
32 #   > NET_INTERFACE=enp2s0          \
33 #     SERVER_ADDRESS='172.24.74.2' \
34 #     SHARED_DIR='/srv/nfs/share'   \
35 #     MOUNT_POINT='/mnt/nfs'
36 #     ./nfs_client_conf
37 #   =====
38
39 #####
40 ### CUSTOM CONFIGURATION PART ###
41 #####
42 NET_INTERFACE="enp2s0"
43 SERVER_ADDRESS="X.X.X.X"
44 SHARED_DIR='path_to_shared_directory'

```

```
45 MOUNT_POINT='path_to_mount_directory'
46
47 #####
48 ### GENERAL CONFIGURATION PART ###
49 #####
50
51 # Set keymap into fr
52 sudo sed -i 's/XKBLAYOUT=".*"/XKBLAYOUT="fr"/' /etc/default/
    keyboard
53
54 # Update the system
55 sudo apt update -y && sudo apt upgrade -y
56
57 # Change the current hostname
58 hostnamectl set-hostname nfs-client
59
60 #####
61 ### NETWORK CONFIGURATION PART ###
62 #####
63
64 # Set dhcp ip address
65 sudo bash -c "cat << 'EOF' > /etc/netplan/01-enp2s0.yaml
66 network:
67   version: 2
68   ethernets:
69     ${NET_INTERFACE}:
70       dhcp4: true
71 EOF"
72
73 # Change 01-enp2s0.yaml accessibility
74 sudo chmod 600 /etc/netplan/01-enp2s0.yaml
75
76 # Apply changes
77 sudo netplan apply
78
79 #####
80 ### NFS-CLIENT CONFIGURATION PART ###
81 #####
82
83 # INFO: To change group name: 'sudo groupmod -n new-name current-
    name'
84 # Install NFS-CLIENT package
85 sudo apt install nfs-common
86
87 # Create mount point
88 sudo mkdir -p ${MOUNT_POINT}
89
90 # Mount the share
91 sudo mount ${SERVER_ADDRESS}:${SHARED_DIR} ${MOUNT_POINT}
92
93 # Verify mount
```



```

94 df -h | grep nfs
95
96 # Umount the share
97 # sudo umount ${MOUNT_POINT}
98
99 # Mount NFS share safely using systemd automount; system boots
   normally even if the NFS server is down.
100 echo "${SERVER_ADDRESS}:${SHARED_DIR} ${MOUNT_POINT} nfs
   defaults,_netdev,x-systemd.automount,nofail 0 0" | sudo tee
   -a /etc/fstab >/dev/null

```

Listing 6: NFS Client Configuration Script

7.8 Screenshots

7.8.1 NFS Test

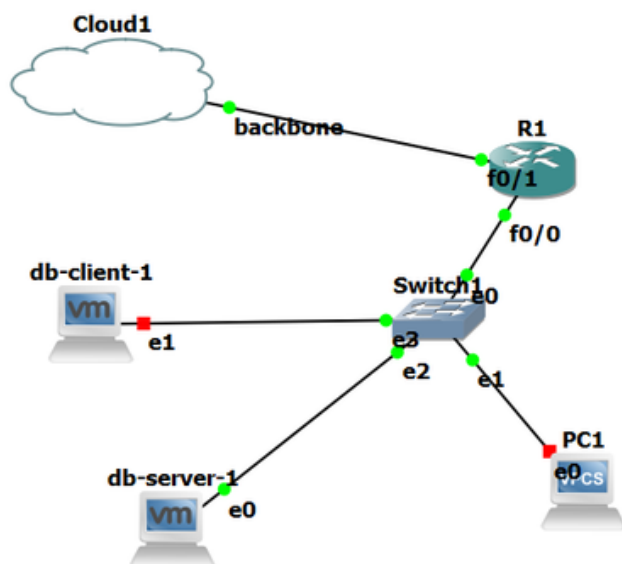
```

zouari_omar@nfs-server:~$ cat /srv/nfs/share/*
Hello Esprit!
Helloooo
zouari_omar@nfs-server:~$ _

```

Figure 2: NFS Test On ‘/srv/nfs/share’

8 Database Department



8.1 Purpose

This department hosts the enterprise MySQL database, centralizing business and client data securely.

8.2 Architecture

- Linux Database Server
- MySQL service
- Restricted client access

8.3 MySQL Configuration

- MySQL bound to internal interface only
- Dedicated database users per service
- Regular backups scheduled

8.4 Security

- ACLs limiting database access
- Strong authentication
- Network isolation via VLAN

8.5 Scripts and Configuration

8.5.1 Database Router Configuration

```
1  ! ZouariOmar <zouariomar20@gmail.com>
2  ! Since:          2025-12-23
3  ! LastUpdate:     2026-01-05
4  ! RouterName:     R-db
5  ! RouterVersion:  C3745
6  ! COPY-THAN-PASTE
7
8  conf t
9  hostname R-db
10
11 ! =====
12 ! ===== INTERFACES PART =====
13 ! === F0/1(WAN) - F0/0(LAN) ===
14 ! =====
15
16 ! === FastEthernet0/1 (WAN) ===
17 interface FastEthernet0/1
18     description R-db to UDP Tunnel / Backbone
19     ip address 10.1.0.2 255.255.255.252
20     no shutdown
21     exit
```

```
22
23 ! === FastEthernet0/0 (LAN) ===
24 interface FastEthernet0/0
25     description R-db to DB-LAN
26     ip address 172.24.72.1 255.255.254.0
27     no shutdown
28     exit
29
30 ! =====
31 ! ===== SERVICES PART =====
32 ! ===== OSPF - DHCP - DNS =====
33 ! =====
34
35 ! === OSPF ===
36 router ospf 1
37     router-id 6.6.6.6
38     log-adjacency-changes
39     network 10.1.0.0 0.0.0.3 area 0
40     network 172.24.72.0 0.0.1.255 area 0
41     passive-interface default
42     no passive-interface FastEthernet0/1
43     default-information originate
44     exit
45
46 ! Default route toward backbone
47 ip route 0.0.0.0 0.0.0.0 10.1.0.1
48
49 ! === DHCP ===
50 ! Excluded addresses
51 ip dhcp excluded-address 172.24.72.1 172.24.72.2
52
53 ip dhcp pool DB-LAN
54     network 172.24.72.0 255.255.254.0
55     default-router 172.24.72.1
56     dns-server 8.8.8.8
57     exit
58
59 ! === DNS ===
60 ip domain-lookup
61 ip name-server 8.8.8.8
62 ip name-server 1.1.1.1
63
64 ! =====
65 ! ===== SECURITY PART =====
66 ! CONSOLE - SSH - TELNET - ACL
67 ! =====
68
69 ! Create Admin user
70 username admin privilege 15 secret <STRONG_PASSWORD>
71
72 ! Encrypt all passwords
```

```
73 service password-encryption
74
75 ! === Console ===
76 line console 0
77     login local
78     exec-timeout 5 0
79     logging synchronous
80     exit
81
82 ! === SSH ===
83 crypto key generate rsa usage-keys label SSH modulus 2048
84 ip ssh version 2
85
86 line vty 0 4
87     login local
88     transport input ssh
89     exec-timeout 5 0
90     logging synchronous
91     exit
92
93 ! === Telnet (optional / legacy) ===
94 line vty 5 15
95     login local
96     transport input telnet
97     exec-timeout 5 0
98     logging synchronous
99     end
100
101 ! Disable unused services
102 no ip http server
103 no ip http secure-server
104 ! no cdp run
105
106 write
107
108 ! =====
109 ! === VERIFICATION PART ===
110 ! =====
111
112 ! === OSPF Interfaces ===
113 ! Check which interfaces run OSPF and their state
114 show ip ospf interface brief
115 show ip ospf interface
116 show ip protocols
117
118 ! === OSPF Interfaces ===
119 ! Verify adjacency with other routers (if backbone is up)
120 show ip ospf neighbor
121 show ip ospf neighbor detail
122
123 ! === OSPF Routes ===
```

```
124 ! Verify OSPF learned routes
125 show ip route ospf
126 show ip route
127
128 ! === Default Route / Redistribution ===
129 ! Check default route advertisement
130 show ip route 0.0.0.0
131 show running-config | include default-information
132
133 ! === LAN Connectivity / DHCP ===
134 ! Verify DHCP bindings and pool
135 show ip dhcp binding
136 show ip dhcp pool
137
138 ! === DNS Resolution ===
139 ! Verify DNS lookup from router
140 ping 8.8.8.8
141 ping www.google.com
```

Listing 7: DB Router Configuration

8.5.2 Database Server Script

```
1 #!/usr/bin/env bash
2 # =====
3 # .FILE
4 #   db_server_conf
5 #
6 # .SYNOPSIS
7 #   Bash script to configure DB-Linux server
8 #
9 # .DESCRIPTION
10 #   I write this script for
11 #   'Linux nfs-server 6.8.0-88-generic #89-Ubuntu SMP
12 #   PREEMPT_DYNAMIC Sat Oct 11 01:02:46 UTC 2025 x86_64 x86_64 GNU
13 #   /Linux'
14 #   (ubuntu-24.04.3-live-server-amd64)
15 #
16 # .PARAMETER NONE
17 #   NONE
18 #
19 # .INPUTS
20 #   NONE
21 #
22 # .OUTPUTS
23 #   NONE
24 #
25 # .NOTES
26 #   Version      : 1.0
27 #   Author       : @ZouariOmar <zouariomar20@gmail.com>
28 #   Created      : 11/22/2025
```

```

27 # Change Log : 9780b36
28 #
29 # .EXAMPLE
30 # > ./db_server_conf
31 # --
32 # > NET_INTERFACE="enp2s0" \
33 # NET_ADDRESS="172.24.74.2/24" \
34 # NET_GW="172.24.74.1" \
35 # NET_DNS_SERVERS="[8.8.8.8]" \
36 # DB_ACCESS_CLIENT="clientdb '@'192.168.2.%'"
37 # ./db_server_conf
38 # =====
39
40 #####
41 ### CUSTOM CONFIGURATION PART ###
42 #####
43 NET_INTERFACE="enp2s0"
44 NET_ADDRESS="X.X.X.X/XX"
45 NET_GW="X.X.X.X"
46 NET_DNS_SERVERS="[X.X.X.X]"
47 DB_ACCESS_CLIENT="'db_client '@'X.X.X.X'"
48
49 #####
50 ### GENERAL CONFIGURATION PART ###
51 #####
52
53 # Set keymap into fr
54 sudo sed -i 's/XKBLAYOUT=".*"/XKBLAYOUT="fr"/' /etc/default/
    keyboard
55
56 # Update the system
57 sudo apt update -y && sudo apt upgrade -y
58
59 # Change the current hostname
60 hostnamectl set-hostname nfs-server
61
62 #####
63 ### NETWORK CONFIGURATION PART ###
64 #####
65
66 # Set the private static ip address
67 sudo bash -c "cat << 'EOF' > /etc/netplan/01-static.yaml
68 network:
69   version: 2
70   renderer: networkd
71   ethernets:
72     ${NET_INTERFACE}:
73       dhcp4: no
74       addresses:
75         - ${NET_ADDRESS}
76       nameservers:

```

```

77     addresses: ${NET_DNS_SERVERS}
78     routes:
79         - to: default
80           via: ${NET_GW}
81 EOF"
82
83 # Change 01-static.yaml accessibility
84 sudo chmod 600 /etc/netplan/01-static.yaml
85
86 # Apply changes
87 sudo netplan apply
88
89 #####
90 ### DB-SERVER CONFIGURATION PART ###
91 #####
92
93 # Install MySQL Server
94 sudo apt install mysql-server -y
95
96 # Enable mysql service
97 sudo systemctl enable --now mysql
98
99 # Allow Remote Connections
100 sudo sed -i 's/^bind-address\s*=\s*/bind-address = 0.0.0.0/' /etc/
    mysql/mysql.conf.d/mysqld.cnf
101 sudo systemctl restart mysql
102
103 # Firewall and Port Check
104 sudo ufw allow 3306/tcp
105 sudo ufw reload
106
107 # Database and User Creation
108 sudo mysql -e "CREATE DATABASE IF NOT EXISTS labdb; CREATE USER
    IF NOT EXISTS ${DB_ACCESS_CLIENT} IDENTIFIED BY 'StrongPass123
    !'; GRANT ALL PRIVILEGES ON labdb.* TO ${DB_ACCESS_CLIENT};
    FLUSH PRIVILEGES;"
109
110 # Create Table and Insert Data
111 sudo mysql -e "USE labdb; CREATE TABLE IF NOT EXISTS users (id
    INT AUTO_INCREMENT PRIMARY KEY, username VARCHAR(50), email
    VARCHAR(100)); INSERT INTO users (username, email) VALUES ('
    server_user1','srv1@mail.com'),('server_user2','srv2@mail.com
    '); SELECT * FROM users;"

```

Listing 8: DB Server Configuration Script

8.5.3 Database Client Script

```

1 #!/usr/bin/env bash
2 # =====
3 # .FILE

```

```

4 # db_client_conf
5 #
6 # .SYNOPSIS
7 #   Bash script to configure DB-Linux client
8 #
9 # .DESCRIPTION
10 #   I write this script for
11 #   'Linux nfs-server 6.8.0-88-generic #89-Ubuntu SMP
12 #   PREEMPT_DYNAMIC Sat Oct 11 01:02:46 UTC 2025 x86_64 x86_64 GNU
13 #   /Linux'
14 #   (ubuntu-24.04.3-live-server-amd64)
15 #
16 # .PARAMETER NONE
17 #   NONE
18 #
19 # .INPUTS
20 #   NONE
21 #
22 # .OUTPUTS
23 #   NONE
24 #
25 # .NOTES
26 #   Version      : 1.0
27 #   Author       : @ZouariOmar <zouariomar20@gmail.com>
28 #   Created      : 11/22/2025
29 #   Change Log   : 9780b36
30 #
31 # .EXAMPLE
32 #   > ./nfs_client_conf
33 #   --
34 #   > NET_INTERFACE=enp2s0 \
35 #       DB_SERVER_IP_ADDRESS="172.24.72.2" \
36 #       DB_CLIENT_NAME="client" \
37 #       DB_CLIENT_PASSWORD="client123"
38 #       ./db_client_conf
39 #   =====
40 #####
41 ### CUSTOM CONFIGURATION PART ###
42 #####
43 NET_INTERFACE="enp2s0"
44 DB_SERVER_IP_ADDRESS="X.X.X.X"
45 DB_CLIENT_NAME="client_name"
46 DB_CLIENT_PASSWORD="client_password"
47 HOSTNAME="db-client"
48 #####
49 ### GENERAL CONFIGURATION PART ###
50 #####
51
52 # Set keymap into fr

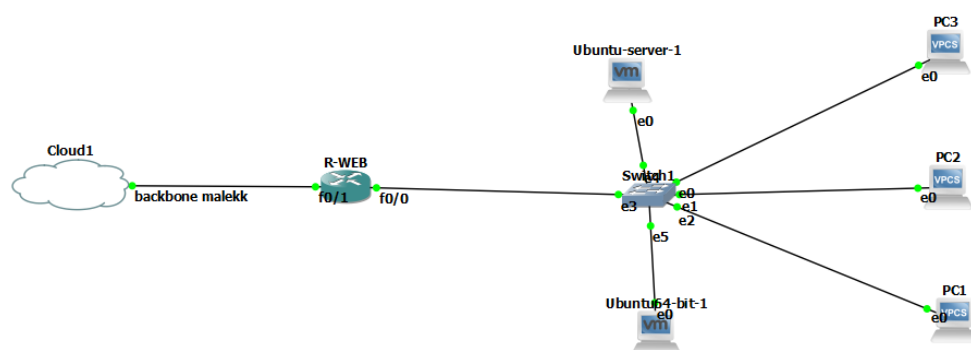
```



```
53 sudo sed -i 's/XKBLAYOUT=".*"/XKBLAYOUT="fr"/' /etc/default/
    keyboard
54
55 # Update the system
56 sudo apt update -y && sudo apt upgrade -y
57
58 # Change the current hostname
59 hostnamectl set-hostname ${HOSTNAME}
60
61 #####
62 ### NETWORK CONFIGURATION PART ###
63 #####
64
65 # Set dhcp ip address
66 sudo bash -c "cat << 'EOF' > /etc/netplan/01-enp2s0.yaml
67 network:
68   version: 2
69   ethernets:
70     ${NET_INTERFACE}:
71       dhcp4: true
72 EOF"
73
74 # Change 01-enp2s0.yaml accessibility
75 sudo chmod 600 /etc/netplan/01-enp2s0.yaml
76
77 # Apply changes
78 sudo netplan apply
79
80 #####
81 ### DB-CLIENT CONFIGURATION PART ###
82 #####
83
84 # Install MySQL Client
85 sudo apt install mysql-client -y
86
87 # Remote Connection && Retrieve Server Data
88 mysql -h ${DB_SERVER_IP_ADDRESS} -u ${DB_CLIENT_NAME} -p -e "SHOW
    DATABASES; USE labdb; SHOW TABLES; SELECT * FROM users;"
```

Listing 9: DB Client Configuration Script

9 Web Department



9.1 Purpose

Hosts the public-facing enterprise website and internal web services.

9.2 Architecture

- Linux Web Server
- Apache2 HTTP server
- Client workstations

9.3 Apache2 Configuration

- Virtual hosts enabled
- Static and dynamic content support
- Logs stored for auditing

9.4 Router and Security

- Port forwarding via NAT
- ACLs permitting HTTP/HTTPS
- OSPF routing integration

9.5 Scripts and Configuration

9.5.1 Web Router Configuration

```

1 ! ZouariOmar <zouariomar20@gmail.com>
2 ! Since:      2025-11-19
3 ! LastUpdate: 2026-01-05
4 ! RouterName: R-WEB

```

```
5  ! RouterVersion: C3745
6  ! COPY-THAN-PASTE
7
8  conf t
9  hostname R-WEB
10
11 ! =====
12 ! ===== INTERFACES PART =====
13 ! === F0/1(WAN) - F0/0(LAN) ===
14 ! =====
15
16 ! === FastEthernet0/1 (WAN) ===
17 interface FastEthernet0/1
18     description R-WEB to UDP Tunnel (RZ-1)
19     ip address 10.3.0.2 255.255.255.252
20     no shutdown
21     exit
22
23 ! === FastEthernet0/0 (LAN) ===
24 interface FastEthernet0/0
25     description LAN WEB-MARKETING
26     ip address 172.24.0.1 255.255.192.0
27     no shutdown
28     exit
29
30 ! Shutdown unused interfaces
31 interface range Serial0/0 , Serial0/1 , Serial0/2 , Serial2/0 -
32     2/3 , FastEthernet1/0
33     shutdown
34     exit
35
36 ! =====
37 ! ===== SERVICES PART =====
38 ! =====
39
40 ! === OSPF ===
41 router ospf 1
42     router-id 10.3.0.2
43     log-adjacency-changes
44     network 10.3.0.0 0.0.0.3 area 0
45     network 172.24.0.0 0.0.63.255 area 0
46     passive-interface default
47     no passive-interface FastEthernet0/1
48     default-information originate
49     exit
50
51 ! Default route toward backbone
52 ip route 0.0.0.0 0.0.0.0 10.3.0.1
53
54 ! === DHCP ===
```

```
55 ip dhcp excluded-address 172.24.0.1 172.24.0.20
56 ip dhcp excluded-address 172.24.63.255
57
58 ip dhcp pool RZ-WEB
59     network 172.24.0.0 255.255.192.0
60     default-router 172.24.0.1
61     dns-server 8.8.8.8
62     domain-name web.techsolutions
63     exit
64
65 ! === DNS ===
66 ip domain-lookup
67 ip name-server 8.8.8.8
68 ip name-server 1.1.1.1
69
70 ! =====
71 ! ===== SECURITY PART =====
72 ! == CONSOLE - SSH - TELNET - ACL ==
73 ! =====
74
75 ! Create an Admin user
76 username admin privilege 15 secret <STRONG_PASSWORD>
77
78 ! Encrypt all passwords
79 service password-encryption
80
81 ! === Console ===
82 line console 0
83     login local
84     exec-timeout 5 0      ! Auto-logout after 5min
85     logging synchronous ! Clean console display
86     exit
87
88 ! === SSH ===
89 crypto key generate rsa usage-keys label SSH modulus 2048
90 ip ssh version 2
91 ! ip access-list standard SSH_ONLY
92 !     permit X.X.X.X X.X.X.X
93 !     deny any
94 line vty 0 4
95     login local
96     transport input ssh
97 ! access-class SSH_ONLY in
98     exec-timeout 5 0
99     logging synchronous
100     exit
101
102 ! === Telnet ===
103 ! ip access-list standard TELNET_ONLY
104 !     permit 10.0.0.0 0.0.0.255
105 !     deny any
```

```
106 line vty 5 15
107     login local
108     transport input telnet
109 ! access-class TELNET_ONLY in
110     exec-timeout 5 0
111     logging synchronous
112     exit
113
114 ! Disable unused services
115 ! no ip http server          ! HTTP
116 ! no ip http secure-server  ! HTTPS
117 ! no cdp run                ! CDP
118
119 write
120
121 ! =====
122 ! === VERIFICATION PART ===
123 ! =====
124
125 ! === OSPF Interfaces ===
126 ! Check which interfaces run OSPF and their state
127 show ip ospf interface brief
128 show ip ospf interface
129 show ip protocols
130
131 ! === OSPF Interfaces ===
132 ! Verify adjacency with other routers (if backbone is up)
133 show ip ospf neighbor
134 show ip ospf neighbor detail
135
136 ! === OSPF Routes ===
137 ! Verify OSPF learned routes
138 show ip route ospf
139 show ip route
140
141 ! === Default Route / Redistribution ===
142 ! Check default route advertisement
143 show ip route 0.0.0.0
144 show running-config | include default-information
145
146 ! === LAN Connectivity / DHCP ===
147 ! Verify DHCP bindings and pool
148 show ip dhcp binding
149 show ip dhcp pool
150
151 ! === DNS Resolution ===
152 ! Verify DNS lookup from router
153 ping 8.8.8.8
154 ping www.google.com
```

Listing 10: Web Router Configuration

9.5.2 Web Server Script

```

1 #!/usr/bin/env bash
2 # =====
3 # .FILE
4 #     web_server_conf
5 #
6 # .SYNOPSIS
7 #     Bash script to configure WEB-Linux server
8 #
9 # .DESCRIPTION
10 #     I write this script for
11 #     'Linux nfs-server 6.8.0-88-generic #89-Ubuntu SMP
12 #     PREEMPT_DYNAMIC Sat Oct 11 01:02:46 UTC 2025 x86_64 x86_64 GNU
13 #     /Linux'
14 #     (ubuntu-24.04.3-live-server-amd64)
15 #
16 # .PARAMETER NONE
17 #     NONE
18 #
19 # .INPUTS
20 #     NONE
21 #
22 # .OUTPUTS
23 #     NONE
24 #
25 # .NOTES
26 #     Version      : 1.0
27 #     Author       : @ZouariOmar <zouariomar20@gmail.com>
28 #     Created      : 11/22/2025
29 #     Change Log   : 9780b36
30 #
31 # .EXAMPLE
32 #     > ./web_server_conf
33 #     --
34 #     > NET_INTERFACE="enp2s0"          \
35 #         NET_ADDRESS="172.24.74.2/24"   \
36 #         NET_GW="172.24.74.1"          \
37 #         NET_DNS_SERVERS="[8.8.8.8]"   \
38 #         WEB_SERVER_ADDRESS='172.24.74.2'
39 #     ./web_server_conf
40 #     =====
41 #####
42 ### CUSTOM CONFIGURATION PART ###
43 #####
44 NET_INTERFACE="enp2s0"
45 NET_ADDRESS="X.X.X.X/XX"
46 NET_GW="X.X.X.X"
47 NET_DNS_SERVERS="[X.X.X.X]"
48 #####

```

```

49 ### GENERAL CONFIGURATION PART ###
50 #####
51
52 # Set keymap into fr
53 sudo sed -i 's/XKBLAYOUT=".*"/XKBLAYOUT="fr"/' /etc/default/
    keyboard
54
55 # Update the system
56 sudo apt update -y && sudo apt upgrade -y
57
58 # Change the current hostname
59 hostnamectl set-hostname web-server
60
61 #####
62 ### NETWORK CONFIGURATION PART ###
63 #####
64
65 # Set the private static ip address
66 sudo bash -c "cat << 'EOF' > /etc/netplan/01-static.yaml
67 network:
68   version: 2
69   renderer: networkd
70   ethernets:
71     ${NET_INTERFACE}:
72       dhcp4: no
73       addresses:
74         - ${NET_ADDRESS}
75       nameservers:
76         addresses: ${NET_DNS_SERVERS}
77       routes:
78         - to: default
79           via: ${NET_GW}
80 EOF"
81
82 # Change 01-static.yaml accessibility
83 sudo chmod 600 /etc/netplan/01-static.yaml
84
85 # Apply changes
86 sudo netplan apply
87
88 #####
89 ### WEB-SERVER CONFIGURATION PART ###
90 #####
91
92 # Install apache2 package
93 sudo apt install apache2 -y
94
95 # Enable apache2 service
96 sudo systemctl enable --now apache2

```

Listing 11: Web Server Configuration Script

9.5.3 Web Client Script

```

1 #!/usr/bin/env bash
2 # =====
3 # .FILE
4 #     web_client_conf
5 #
6 # .SYNOPSIS
7 #     Bash script to configure WEB-Linux client
8 #
9 # .DESCRIPTION
10 #     I write this script for
11 #     'Linux nfs-server 6.8.0-88-generic #89-Ubuntu SMP
12 #     PREEMPT_DYNAMIC Sat Oct 11 01:02:46 UTC 2025 x86_64 x86_64 GNU
13 #     /Linux'
14 #     (ubuntu-24.04.3-live-server-amd64)
15 #
16 # .PARAMETER NONE
17 #     NONE
18 #
19 # .INPUTS
20 #     NONE
21 #
22 # .OUTPUTS
23 #     NONE
24 #
25 # .NOTES
26 #     Version      : 1.0
27 #     Author       : @ZouariOmar <zouariomar20@gmail.com>
28 #     Created      : 11/22/2025
29 #     Change Log   : 9780b36
30 #
31 # .EXAMPLE
32 #     > ./web_client_conf
33 #     --
34 #     > NET_INTERFACE=enp2s0 \
35 #     WEB_SERVER_ADDRESS='172.24.74.2'
36 #     ./web_client_conf
37 #     =====
38 #####
39 ### CUSTOM CONFIGURATION PART ###
40 #####
41 NET_INTERFACE="enp2s0"
42 WEB_SERVER_ADDRESS="X.X.X.X"
43 #####
44 ### GENERAL CONFIGURATION PART ###
45 #####
46
47 # Set keymap into fr

```



```
48 sudo sed -i 's/XKBLAYOUT=".*"/XKBLAYOUT="fr"/' /etc/default/
    keyboard
49
50 # Update the system
51 sudo apt update -y && sudo apt upgrade -y
52
53 # Change the current hostname
54 hostnamectl set-hostname web-client
55
56 #####
57 ### NETWORK CONFIGURATION PART ###
58 #####
59
60 # Set dhcp ip address
61 sudo bash -c "cat << 'EOF' > /etc/netplan/01-enp2s0.yaml
62 network:
63     version: 2
64     ethernets:
65         ${NET_INTERFACE}:
66             dhcp4: true
67 EOF"
68
69 # Change 01-enp2s0.yaml accessibility
70 sudo chmod 600 /etc/netplan/01-enp2s0.yaml
71
72 # Apply changes
73 sudo netplan apply
74
75 #####
76 ### WEB-CLIENT CONFIGURATION PART ###
77 #####
78
79 # Test the Web server
80 curl ${WEB_SERVER_ADDRESS}
```

Listing 12: Web Client Configuration Script

9.6 Screenshots

9.6.1 Web Client Interface

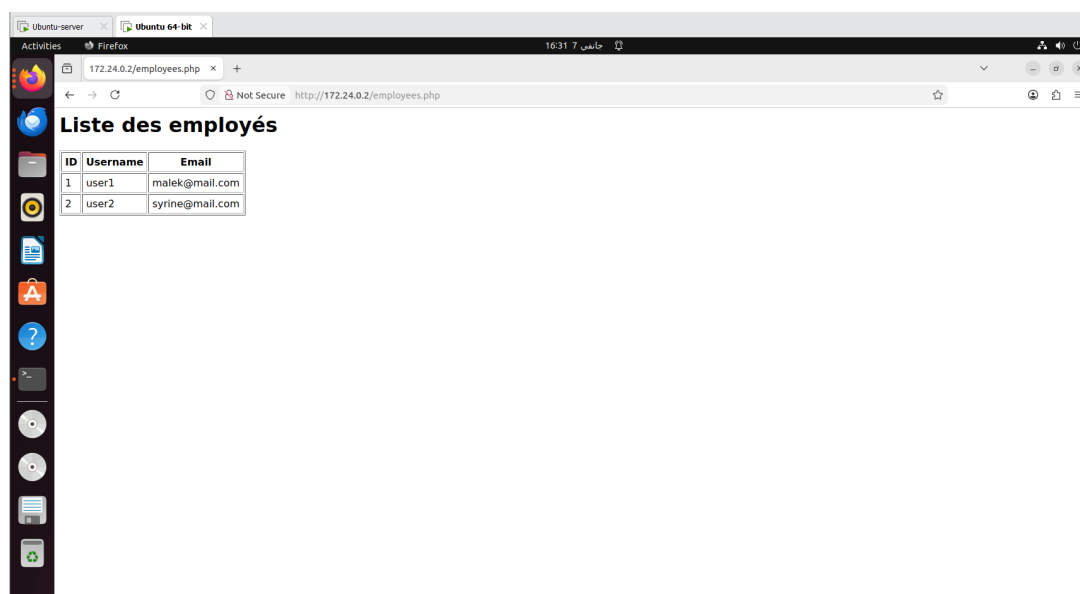
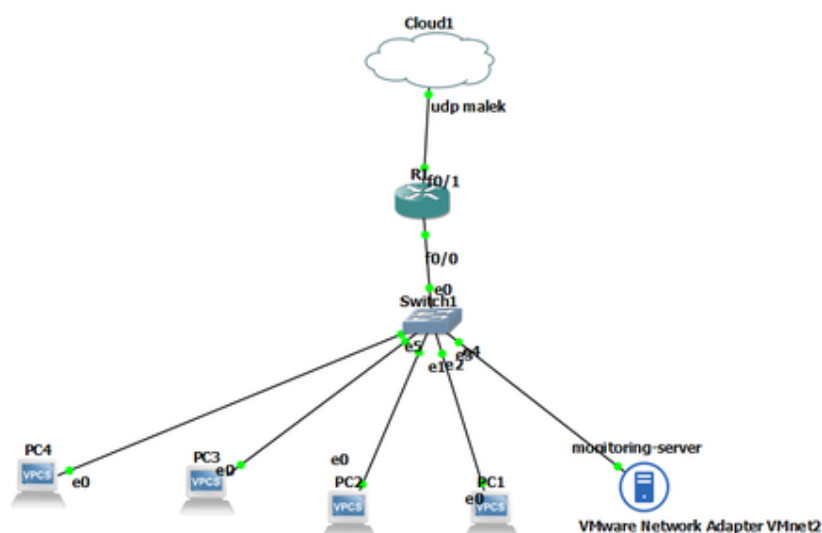


Figure 3: Client that access to 'employee.php' page

10 Monitoring Department



10.1 Purpose

Provides real-time monitoring, metrics collection, and visualization of the entire infrastructure.

10.2 Architecture

- Linux Monitoring Server
- Prometheus for metrics collection
- Grafana for visualization

10.3 Prometheus

- Scrapes routers and servers
- Node exporter installed on Linux hosts
- Custom alert rules defined

10.4 Grafana

- Dashboards for CPU, RAM, network usage
- Role-based access control
- Historical data analysis

10.5 Scripts and Configuration

10.5.1 Monitoring Router Configuration

```
1 ! ZouariOmar <zouariomar20@gmail.com>
2 ! Since:          2025-11-19
3 ! LastUpdate:     2026-01-05
4 ! RouterName:     R-MONITORING
5 ! RouterVersion:  C3745
6 ! COPY-THAN-PASTE
7
8 conf t
9 hostname R-MONITORING
10
11 ! =====
12 ! ===== INTERFACES PART =====
13 ! === FO/1(WAN) - FO/0(LAN) ===
14 ! =====
15
16 ! === FastEthernet0/1 (WAN) ===
17 interface FastEthernet0/1
18     description R-MONITORING to UDP Tunnel (RZ-1)
19     ip address 10.3.0.6 255.255.255.252
20     no shutdown
21     exit
22
23 ! === FastEthernet0/0 (LAN) ===
24 interface FastEthernet0/0
25     description LAN MONITORING
26     ip address 172.24.64.1 255.255.192.0
```

```
27  no shutdown
28  exit
29
30  ! Shutdown unused interfaces
31  interface range Serial0/0 , Serial0/1 , Serial0/2 , Serial2/0 -
    2/3 , FastEthernet1/0
32  shutdown
33  exit
34
35  ! =====
36  ! ===== SERVICES PART =====
37  ! ===== OSPF - DHCP - DNS =====
38  ! =====
39
40  ! === OSPF ===
41  router ospf 1
42  router-id 10.3.0.6
43  log-adjacency-changes
44  network 10.3.0.4 0.0.0.3 area 0
45  network 172.24.64.0 0.0.63.255 area 0
46  passive-interface default
47  no passive-interface FastEthernet0/1
48  default-information originate
49  exit
50
51  ! Default route toward backbone
52  ip route 0.0.0.0 0.0.0.0 10.3.0.5
53
54  ! === DHCP ===
55  ip dhcp excluded-address 172.24.64.1 172.24.64.20
56  ip dhcp excluded-address 172.24.127.255
57
58  ip dhcp pool RZ-MONITORING
59  network 172.24.64.0 255.255.192.0
60  default-router 172.24.64.1
61  dns-server 8.8.8.8
62  domain-name monitoring.techsolutions
63  exit
64
65  ! === DNS ===
66  ip domain-lookup
67  ip name-server 8.8.8.8
68  ip name-server 1.1.1.1
69
70  ! =====
71  ! ===== SECURITY PART =====
72  ! == CONSOLE - SSH - TELNET - ACL ==
73  ! =====
74
75  ! Create an Admin user
76  username admin privilege 15 secret <STRONG_PASSWORD>
```

```
77
78 ! Encrypt all passwords
79 service password-encryption
80
81 ! === Console ===
82 line console 0
83     login local
84     exec-timeout 5 0
85     logging synchronous
86     exit
87
88 ! === SSH ===
89 crypto key generate rsa usage-keys label SSH modulus 2048
90 ip ssh version 2
91 line vty 0 4
92     login local
93     transport input ssh
94     exec-timeout 5 0
95     logging synchronous
96     exit
97
98 ! === Telnet ===
99 line vty 5 15
100     login local
101     transport input telnet
102     exec-timeout 5 0
103     logging synchronous
104     exit
105
106 ! Disable unused services
107 ! no ip http server
108 ! no ip http secure-server
109 ! no cdp run
110
111 write
112
113 ! =====
114 ! === VERIFICATION PART ===
115 ! =====
116
117 ! === OSPF Interfaces ===
118 show ip ospf interface brief
119 show ip ospf interface
120 show ip protocols
121
122 ! === OSPF Interfaces ===
123 show ip ospf neighbor
124 show ip ospf neighbor detail
125
126 ! === OSPF Routes ===
127 show ip route ospf
```

```
128 show ip route
129
130 ! === Default Route / Redistribution ===
131 show ip route 0.0.0.0
132 show running-config | include default-information
133
134 ! === LAN Connectivity / DHCP ===
135 show ip dhcp binding
136 show ip dhcp pool
137
138 ! === DNS Resolution ===
139 ping 8.8.8.8
140 ping www.google.com
```

Listing 13: Monitoring Router Configuration

10.5.2 Monitoring Server Script

```
1 #!/usr/bin/env bash
2 # =====
3 # .FILE
4 #   monitoring_server_conf
5 #
6 # .SYNOPSIS
7 #   Bash script to configure Prometheus,
8 #   Node Exporter, and Grafana monitoring stack
9 #
10 # .DESCRIPTION
11 #   This script installs and configures:
12 #       - Prometheus (systemd service)
13 #       - Node Exporter
14 #       - Grafana
15 #
16 #   Tested on:
17 #   Linux 6.8.0-88-generic #89-Ubuntu SMP PREEMPT_DYNAMIC
18 #   (ubuntu-24.04.3-live-server-amd64)
19 #
20 # .PARAMETER NONE
21 #   NONE
22 #
23 # .INPUTS
24 #   NONE
25 #
26 # .OUTPUTS
27 #   NONE
28 #
29 # .NOTES
30 #   Version      : 1.0
31 #   Author       : @ZouariOmar <zouariomar20@gmail.com>
32 #   Created      : 01/07/2026
33 #   Change Log   : initial-release
```

```
34 #
35 # .EXAMPLE
36 #   > ./monitoring_server_conf
37 #
38 # =====
39
40 #####
41 ### CUSTOM CONFIGURATION PART ###
42 #####
43
44 PROM_VERSION="1.8.1"
45 PROM_USER="prometheus"
46 NODE_USER="node_exporter"
47
48 PROM_DIR="/etc/prometheus"
49 PROM_DATA_DIR="/var/lib/prometheus"
50 PROM_BIN="/usr/local/bin/prometheus"
51
52 #####
53 ### GENERAL CONFIGURATION PART ###
54 #####
55
56 # Update system
57 sudo apt update -y && sudo apt upgrade -y
58
59 # Install required packages
60 sudo apt install -y wget curl tar apt-transport-https software-
    properties-common
61
62 #####
63 ### PROMETHEUS CONFIGURATION ###
64 #####
65
66 # Create Prometheus user
67 sudo useradd --no-create-home --shell /bin/false ${PROM_USER}
68
69 # Create directories
70 sudo mkdir -p ${PROM_DIR}
71 sudo mkdir -p ${PROM_DATA_DIR}
72
73 sudo chown ${PROM_USER}:${PROM_USER} ${PROM_DIR}
74 sudo chown ${PROM_USER}:${PROM_USER} ${PROM_DATA_DIR}
75
76 # Download Prometheus
77 cd /tmp || exit
78 wget https://github.com/prometheus/prometheus/releases/download/
    v${PROM_VERSION}/prometheus-${PROM_VERSION}.linux-amd64.tar.gz
79
80 # Extract
81 tar xvf prometheus-${PROM_VERSION}.linux-amd64.tar.gz
82
```

```
83 # Install binaries
84 sudo cp prometheus-${PROM_VERSION}.linux-amd64/prometheus /usr/
    local/bin/
85 sudo cp prometheus-${PROM_VERSION}.linux-amd64/promtool /usr/
    local/bin/
86
87 sudo chown ${PROM_USER}:${PROM_USER} /usr/local/bin/prometheus
88 sudo chown ${PROM_USER}:${PROM_USER} /usr/local/bin/promtool
89
90 #####
91 ### PROMETHEUS CONFIG FILE ###
92 #####
93
94 sudo bash -c "cat << 'EOF' > ${PROM_DIR}/prometheus.yml
95 global:
96     scrape_interval: 15s
97
98 scrape_configs:
99     - job_name: 'prometheus'
100       static_configs:
101         - targets: ['localhost:9090']
102
103     - job_name: 'node_exporter'
104       static_configs:
105         - targets: ['localhost:9100']
106 EOF"
107
108 sudo chown ${PROM_USER}:${PROM_USER} ${PROM_DIR}/prometheus.yml
109
110 #####
111 ### PROMETHEUS SYSTEMD SERVICE ###
112 #####
113
114 sudo bash -c "cat << 'EOF' > /etc/systemd/system/prometheus.
    service
115 [Unit]
116 Description=Prometheus
117 After=network-online.target
118
119 [Service]
120 User=${PROM_USER}
121 ExecStart=${PROM_BIN} \
122     --config.file=${PROM_DIR}/prometheus.yml \
123     --storage.tsdb.path=${PROM_DATA_DIR}
124
125 [Install]
126 WantedBy=multi-user.target
127 EOF"
128
129 # Enable and start Prometheus
130 sudo systemctl daemon-reload
```



```
131 sudo systemctl enable --now prometheus
132
133 #####
134 ### NODE EXPORTER CONFIGURATION ###
135 #####
136
137 # Create Node Exporter user
138 sudo useradd --no-create-home --shell /bin/false ${NODE_USER}
139
140 # Download Node Exporter
141 cd /tmp || exit
142 wget https://github.com/prometheus/node_exporter/releases/
    download/v${PROM_VERSION}/node_exporter-${PROM_VERSION}.linux-
    amd64.tar.gz
143
144 # Extract
145 tar xvf node_exporter-${PROM_VERSION}.linux-amd64.tar.gz
146
147 # Install binary
148 sudo cp node_exporter-${PROM_VERSION}.linux-amd64/node_exporter /
    usr/local/bin/
149 sudo chown ${NODE_USER}:${NODE_USER} /usr/local/bin/node_exporter
150
151 #####
152 ### NODE EXPORTER SYSTEMD ###
153 #####
154
155 sudo bash -c "cat << 'EOF' > /etc/systemd/system/node_exporter.
    service
156 [Unit]
157 Description=Node Exporter
158 After=network-online.target
159
160 [Service]
161 User=${NODE_USER}
162 ExecStart=/usr/local/bin/node_exporter
163
164 [Install]
165 WantedBy=multi-user.target
166 EOF"
167
168 # Enable and start Node Exporter
169 sudo systemctl daemon-reload
170 sudo systemctl enable --now node_exporter
171
172 #####
173 ### GRAFANA INSTALLATION ###
174 #####
175
176 # Add Grafana repository
```

```

177 wget -q -O - https://packages.grafana.com/gpg.key | sudo apt-key
    add -
178 sudo add-apt-repository "deb https://packages.grafana.com/oss/deb
    stable main"
179
180 # Install Grafana
181 sudo apt update
182 sudo apt install -y grafana
183
184 # Enable and start Grafana
185 sudo systemctl enable --now grafana-server
186
187 #####
188 ### VERIFICATION PART ###
189 #####
190
191 echo "===== "
192 echo " Monitoring stack installation completed "
193 echo "===== "
194 echo " Prometheus : http://localhost:9090"
195 echo " Grafana      : http://localhost:3000"
196 echo " Default Grafana credentials:"
197 echo "   user: admin"
198 echo "   pass: admin"
199 echo "===== "

```

Listing 14: Monitoring Server Configuration Script

10.5.3 Monitoring Client Script

```

1  #!/usr/bin/env bash
2  # =====
3  # .FILE
4  #   monitoring_client_conf
5  #
6  # .SYNOPSIS
7  #   Bash script to configure MONITORING-Linux client
8  #
9  # .DESCRIPTION
10 #   I write this script for
11 #   'Linux nfs-server 6.8.0-88-generic #89-Ubuntu SMP
    PREEMPT_DYNAMIC Sat Oct 11 01:02:46 UTC 2025 x86_64 x86_64 GNU
    /Linux '
12 #   (ubuntu-24.04.3-live-server-amd64)
13 #
14 # .PARAMETER NONE
15 #   NONE
16 #
17 # .INPUTS
18 #   NONE
19 #

```

```

20 # .OUTPUTS
21 #     NONE
22 #
23 # .NOTES
24 #     Version      : 1.0
25 #     Author       : @ZouariOmar <zouariomar20@gmail.com>
26 #     Created      : 01/07/2026
27 #     Change Log   : 9780b36
28 #
29 # .EXAMPLE
30 #     > ./nfs_client_conf
31 #     --
32 #     > NET_INTERFACE=enp2s0 ./monitoring_client_conf
33 #     =====
34
35 #####
36 ### CUSTOM CONFIGURATION PART ###
37 #####
38 NET_INTERFACE="enp2s0"
39 HOSTNAME="monitoring-client"
40
41 #####
42 ### GENERAL CONFIGURATION PART ###
43 #####
44
45 # Set keymap into fr
46 sudo sed -i 's/XKBLAYOUT=".*"/XKBLAYOUT="fr"/' /etc/default/
    keyboard
47
48 # Update the system
49 sudo apt update -y && sudo apt upgrade -y
50
51 # Change the current hostname
52 hostnamectl set-hostname ${HOSTNAME}
53
54 #####
55 ### NETWORK CONFIGURATION PART ###
56 #####
57
58 # Set dhcp ip address
59 sudo bash -c "cat << 'EOF' > /etc/netplan/01-enp2s0.yaml
60 network:
61     version: 2
62     ethernets:
63         ${NET_INTERFACE}:
64             dhcp4: true
65 EOF"
66
67 # Change 01-enp2s0.yaml accessibility
68 sudo chmod 600 /etc/netplan/01-enp2s0.yaml
69

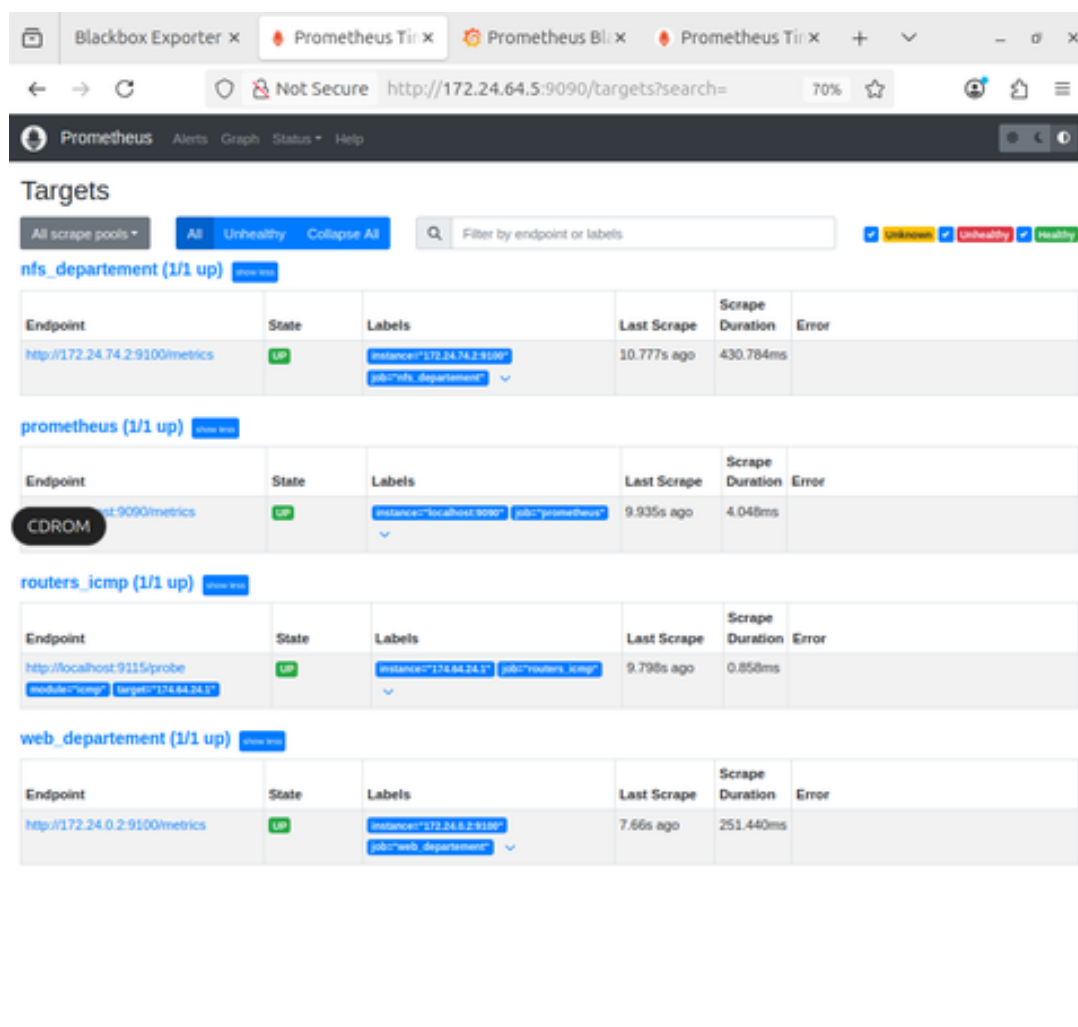
```

```
70 # Apply changes
71 sudo netplan apply
72
73 #####
74 ### MONITORING-CLIENT CONFIGURATION PART ###
75 #####
76
77 # Download node_exporter archive to /tmp
78 cd /tmp && wget https://github.com/prometheus/node_exporter/
    releases/download/v1.6.1/node_exporter-1.6.1.linux-amd64.tar.
    gz
79
80 # Extract the archive
81 tar xvf node_exporter-1.6.1.linux-amd64.tar.gz
82
83 # Move the node_exporter binary to a system-wide location
84 sudo mv node_exporter-1.6.1.linux-amd64/node_exporter /usr/local/
    bin/
85
86 # Create a dedicated non-login system user for security
87 sudo useradd -rs /bin/false node_exporter
88
89 # Create systemd service file (no nano, written inline)
90 sudo tee /etc/systemd/system/node_exporter.service >/dev/null <<'
    EOF'
91 [Unit]
92 Description=Node Exporter
93 After=network.target
94
95 [Service]
96 User=node_exporter
97 ExecStart=/usr/local/bin/node_exporter
98
99 [Install]
100 WantedBy=multi-user.target
101 EOF
102
103 # Reload systemd to pick up the new service
104 sudo systemctl daemon-reload
105
106 # Enable node_exporter at boot and start it immediately
107 sudo systemctl enable --now node_exporter
108
109 # Verify node_exporter is exposing metrics locally
110 # You should see Prometheus-style metrics output
111 curl http://localhost:9100/metrics
```

Listing 15: Monitoring Client Configuration Script

10.6 Screenshots

10.6.1 Prometheus Web Interface



The screenshot displays the Prometheus Web Interface in a browser window. The address bar shows the URL `http://172.24.64.5:9090/targets?search=`. The interface includes a navigation bar with links for Alerts, Graph, Status, and Help. The main section is titled "Targets" and features a search bar and filters for "All scrape pools", "All", "Unhealthy", and "Collapse All". Below this, there are four sections, each representing a different target group: `nfs_departement`, `prometheus`, `routers_icmp`, and `web_departement`. Each section contains a table with columns for Endpoint, State, Labels, Last Scrape, Scrape Duration, and Error. All targets are currently in a "UP" state.

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
nfs_departement (1/1 up)					
<code>http://172.24.74.2:9100/metrics</code>	UP	<code>instance="172.24.74.2:9100"</code> <code>job="nfs_departement"</code>	10.777s ago	430.784ms	
prometheus (1/1 up)					
<code>http://localhost:9090/metrics</code>	UP	<code>instance="localhost:9090"</code> <code>job="prometheus"</code>	9.935s ago	4.048ms	
routers_icmp (1/1 up)					
<code>http://localhost:9115/probe</code>	UP	<code>instance="174.64.24.1"</code> <code>job="routers_icmp"</code> <code>module="icmp"</code> <code>target="174.64.24.1"</code>	9.798s ago	0.858ms	
web_departement (1/1 up)					
<code>http://172.24.0.2:9100/metrics</code>	UP	<code>instance="172.24.0.2:9100"</code> <code>job="web_departement"</code>	7.66s ago	251.440ms	

Figure 4: Prometheus Web Interface Showing Active Targets

10.6.2 Grafana Dashboard

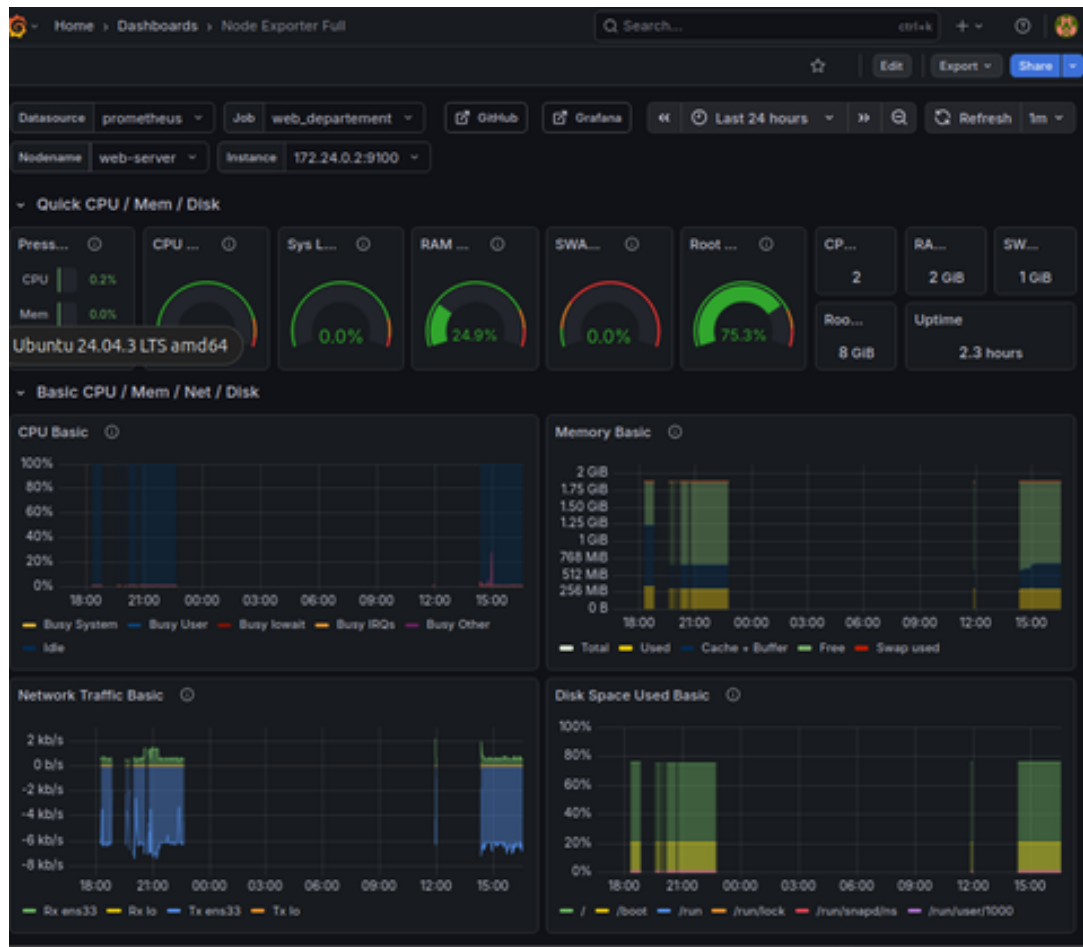


Figure 5: Grafana Node Exporter Dashboard (ID: 1860)

11 Conclusion

This document summarizes the complete network architecture for Tech Solutions (SARL). The simulated environment provides a realistic, scalable, and secure platform for training, experimentation, and enterprise design validation.

End of Report

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